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Ketones in Urine

What is a ketones in urine test?

A ketones in urine test measures ketone levels in your urine. Normally, the cells in your body use **blood glucose** [<https://medlineplus.gov/bloodglucose.html>] (also called blood sugar) for energy. It comes from the food you eat. Your body breaks down most of that food into glucose and releases it into your bloodstream. If your cells can't get enough glucose, your body breaks down fat for energy instead. This produces an acid called ketones, which can build up in your blood and urine.

Having some ketones in your urine is normal. But high ketone levels in your urine may be a sign that your blood is too acidic. This condition is called ketoacidosis. The most common type of ketoacidosis is a **complication of diabetes** [<https://medlineplus.gov/diabetescomplications.html>] called diabetic ketoacidosis (DKA). Diabetic ketoacidosis is a medical emergency that often develops quickly and can be life-threatening.

If you have diabetes, a ketones in urine test can help find high ketone levels early, so you can get treatment right away. **Home tests** [<https://medlineplus.gov/lab-tests/at-home-medical-tests/>] are available if your health care provider recommends frequent testing.

Other names: ketones urine test, ketone test, urine ketones, ketone bodies

What is it used for?

A ketones in urine test is often used to help monitor people who are more likely to develop high levels of ketones. You are more at risk if you have a medical condition, diet, or exercise habits that cause your body to use fat for energy instead of glucose, such as:

- **Diabetes** [<https://medlineplus.gov/diabetes.html>] . Diabetic ketoacidosis (DKA) is a life-threatening condition caused by high ketone levels. It's most common in people who have **type 1 diabetes** [<https://medlineplus.gov/diabetestype1.html>] . But people who have **type 2 diabetes**

[<https://medlineplus.gov/diabetestype2.html>] can also develop DKA.

- Vomiting [<https://medlineplus.gov/nauseaandvomiting.html>] or diarrhea [<https://medlineplus.gov/diarrhea.html>] that lasts a long time.
- Alcohol use disorder (AUD) [<https://medlineplus.gov/alcoholusedisorderaud.html>] , which can cause alcoholic ketoacidosis.
- Eating disorders [<https://medlineplus.gov/eatingdisorders.html>] and starvation.
- Certain digestive diseases [<https://medlineplus.gov/digestivediseases.html>] .
- **Eating a "keto diet,"** which is low in carbohydrates [<https://medlineplus.gov/carbohydrates.html>] (foods that supply glucose for energy).
- **Exercising** too hard for a long time.
- **Pregnancy.**

Why do I need a ketones in urine test?

Your provider may order a ketones in urine test if you have diabetes or other conditions that increase your risk for developing high ketone levels. You may also need this test if you have symptoms of ketoacidosis, which may include:

Early symptoms of ketoacidosis:

- Urinating (peeing) a lot
- Extreme hunger or thirst
- Dehydration [<https://medlineplus.gov/dehydration.html>]
- Headache [<https://medlineplus.gov/headache.html>]

Severe symptoms of ketoacidosis:

- Nausea or vomiting
- Pain in your abdomen [<https://medlineplus.gov/abdominalpain.html>] (belly)
- Confusion
- Trouble breathing
- Feeling very tired or sleepy
- Fruity-smelling breath

If you have severe symptoms of ketoacidosis, seek immediate medical help. Large amounts of ketones can be life-threatening.

What happens during a ketones in urine test?

A ketones in urine test can be done at home or in a lab. If you go to a lab for your test, you will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

1. Wash your hands with soap and water and dry them.
2. Open the container without touching the inside.
3. Clean your genital area with the cleansing wipe:
 - For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.
 - For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.
4. Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.
5. Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.
6. Finish urinating into the toilet.
7. Put the cap on the container and return it as instructed.

If you have hemorrhoids [<https://medlineplus.gov/hemorrhoids.html>] that bleed or are having your menstrual period [<https://medlineplus.gov/menstruation.html>] , tell your provider before your test.

If you do the test at home, follow the instructions in your test kit. Your kit will include a package of test strips. You will either hold a test strip in your urine stream or dip it into a cup of urine that you collect with the clean-catch method. It's important to follow the test instructions exactly. Talk with your provider if you have any questions about how you should test for ketones in urine.

Will I need to do anything to prepare for the test?

You may have to fast (not eat or drink) for a certain period of time before taking a ketones in urine test. Ask your provider if you need to fast or do any other type of preparation before your test.

Are there any risks to the test?

There is no known risk to having a ketones in urine test.

What do the results mean?

Your test results may be a specific number or listed as a "small," "moderate," or "large" amount of ketones. Normal results can vary depending on your diet, activity level, and other factors.

Because **high ketone levels** can cause serious health problems, be sure to talk with your provider about what is normal for you and what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results [<https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/>] .

Is there anything else I need to know about a ketones in urine test?

Ketone test kits are available at most pharmacies without a prescription. If you are planning to test for ketones at home, ask your provider which kit would be best for you. At-home urine tests are easy to do and can provide accurate results as long as you carefully follow all the instructions.

Some people use at-home kits to test for ketones if they are on a ketogenic or "keto" diet to try to lose weight. On a keto diet, you eat mainly foods with little or no carbohydrates. This makes your body use fat instead of glucose for energy, which may increase your ketone levels. A keto diet may not be right for people with certain conditions. Talk with your provider before going on a keto diet to see if it is right for you.

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